

INTERNSHIP PROJECT- WEEK 1

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Project Title: House Price Prediction

Problem Statement:

Real estate buyers and sellers often rely on guesswork or outdated comparisons to estimate a property's fair value. Your task is to build a regression model that predicts house prices based on property features such as size, number of rooms, location, and age — and then identify which features most strongly influence price.

Summary:

This project involved analysing a housing dataset to predict house prices with machine learning techniques. First, we loaded the dataset and explored it with Pandas. We looked at its structure, the number of rows and columns, and any missing values. We identified the target variable as Price. The other columns, like area, number of bedrooms, bathrooms, stories, mainroad, guestroom, basement, hotwaterheating, airconditioning, parking, prefarea, furnishingstatus, were treated as features. We cleaned the data by addressing missing values, removing duplicates, and converting categorical variables into numbers using one-hot encoding. This made the data ready for machine learning models. After preprocessing, we split the dataset into training and testing sets, using an 80/20 ratio. We built two regression models: Linear Regression and Random Forest Regressor. We trained both models to predict house prices and evaluated them using MAE, RMSE, and R^2 score. The Random Forest model performed better overall because it captured more complex and non-linear patterns in the data compared to Linear Regression, which served as our baseline model. We also used visualization to better understand the dataset. A histogram displayed the distribution of house prices, a correlation heatmap pointed out the strongest relationships between features and price, and a scatter plot of my choice compared actual vs predicted values.

The analysis revealed that area, bathrooms, and certain location-related features had the strongest impact on house prices. In conclusion, the study shows that machine learning models can effectively predict house prices based on property features. A real estate business can use these models to estimate property values more accurately and focus on key features like location, size, and amenities when making pricing or investment decisions.